

Weakly Supervised Video Anomaly Detection Using Spatiotemporal MIL, Temporal Attention and CLIP-Based Semantic Alignment

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Abstract—Video Anomaly Detection (VAD) aims to identify abnormal events in long surveillance videos, but frame-level annotation is expensive and difficult to scale. This project implements a weakly supervised Multiple Instance Learning (MIL) framework using video-level labels and precomputed temporal feature bags. A Sultani-style ranking baseline is first developed using fixed-length temporal feature bags and then extended with a temporal self-attention MIL model that learns relationships between video segments. UCF-Crime is the main surveillance dataset and is represented using R(2+1)D clip-level features, while XD-Violence support is included through available I3D RGB and optical-flow feature bags. Attention MIL improves test video-level AUC from 0.9312 to 0.9498 and test Average Precision (AP) from 0.9534 to 0.9630 over the baseline. An extra-credit CLIP semantic head further raises test AUC/AP to 0.9765/0.9846, but its fine-grained semantic macro-F1 remains low at 0.2390, showing that semantic category recognition remains difficult under weak video-level supervision. The final results show that temporal attention is the main practical improvement, while the CLIP head is best treated as an interpretability extension rather than a fully reliable fine-grained anomaly classifier.

Index Terms—Video anomaly detection, weakly supervised learning, multiple instance learning, temporal attention, CLIP, UCF-Crime, XD-Violence.

I. INTRODUCTION

Video surveillance is widely used in public spaces, transport hubs, campuses and private property. However, manual monitoring of long video streams is time-consuming and prone to human error. Video Anomaly Detection (VAD) aims to automatically identify unusual events such as violence, road accidents, explosions, robbery and other abnormal behaviour. The main challenge is supervision: fully supervised methods require precise frame-level temporal annotations, while real-world datasets often provide only video-level labels.

Weakly supervised VAD addresses this problem by treating each video as a bag of temporal segments. A normal video is assumed to contain only normal segments, while an anomalous video contains at least one abnormal segment. This naturally leads to a Multiple Instance Learning (MIL) formulation, where segment-level anomaly scores are learned from video-level labels only.

This project follows the MIL formulation introduced by Sultani et al. [2]. A baseline model scores each segment independently using a shared multilayer perceptron and is trained using a ranking loss. The main improvement is a temporal self-attention MIL model inspired by Transformer attention [9], allowing each segment to be scored in relation to other segments in the same video. Finally, a CLIP-based semantic head is added as an extra-credit extension to provide text-label suggestions for anomaly categories.

The main contributions are:

- A complete weakly supervised VAD pipeline using temporal feature bags from UCF-Crime and available XD-Violence RGB/Flow features.
- A Sultani-style MIL baseline with ranking loss, sparsity regularisation and top- K pooling.
- A temporal-attention MIL model that improves validation and test AUC/AP over the baseline.
- A structured experimental comparison covering baseline MIL, attention MIL, pooling/hyperparameter ablations and CLIP semantic alignment.
- Qualitative analysis using UMAP, anomaly score curves, attention heatmaps and CLIP semantic confusion matrices.

II. RELATED WORK

A. Unsupervised Video Anomaly Detection

Early VAD approaches often model normal behaviour and treat deviations from normality as anomalies. Hasan et al. [1] used autoencoders to learn temporal regularity in video sequences, where abnormal events produce higher reconstruction error. Although such approaches are useful when normal behaviour is consistent, they can struggle when normal scenes are diverse or when anomalous and normal motion patterns overlap.

B. Weakly Supervised MIL for VAD

Sultani et al. [2] introduced UCF-Crime and formulated anomaly detection as a weakly supervised MIL problem. In this setup, each video is a bag and each temporal segment is an instance. Their ranking loss encourages the highest-scoring segment in an anomalous video to exceed the highest-scoring segment in a normal video. Sparsity and temporal smoothness are used to discourage assigning high anomaly scores to all segments.

RTFM [3] further showed that abnormal snippets often have stronger temporal feature magnitude than normal snippets. This supports the use of top- K aggregation instead of pure max pooling, because max pooling can be dominated by a single noisy segment.

C. XD-Violence and Multimodal Cues

Wu et al. [4] introduced XD-Violence, a large-scale audio-visual violence dataset. Their work shows that visual and audio streams can provide complementary information for violent-event detection. In this project, XD-Violence support is included through available RGB and optical-flow visual feature bags, while audio is left for future work.

D. Context Bias in Weak Supervision

Lv et al. [5] identify a key problem in weakly supervised MIL: models can become biased toward context shortcuts rather than true anomaly cues. For example, smoke, sudden camera motion or unusual lighting may be incorrectly treated as evidence of abnormality. This is relevant to the overlap in UMAP visualisations and the false-positive score curves observed in this project.

E. CLIP for Semantic Alignment

CLIP [6] learns a shared embedding space between images and text. Because it can encode natural-language class names, it is useful for zero-shot recognition and semantic alignment. In this project, a frozen CLIP text encoder provides semantic prototypes for anomaly labels, while the video model learns a projection into CLIP text space.

TABLE I: Dataset and Feature Configuration

Dataset	Features	Temporal Cue	Use
UCF-Crime	R(2+1)D	Spatiotemporal	Main VAD data
XD-Violence	I3D RGB + Flow	Visual motion	Feature integration
XD audio	Not used	Audio omitted	Future work

TABLE II: Final Manifest Summary

Split	UCF-Crime Clips	XD Feature Bags	Combined Total
Train	1420	Available paired bags	1421
Validation	190	Available paired bags	191
Test	290	Available paired bags	292

The final quantitative conclusions are primarily driven by UCF-Crime, while XD-Violence is included through the available paired RGB/Flow feature-bag integration.

III. METHODOLOGY

A. Dataset and Feature Representation

The pipeline uses UCF-Crime as the main surveillance anomaly dataset. UCF-Crime is represented using R(2+1)D spatiotemporal clip-level features. XD-Violence integration is included using available I3D RGB and optical-flow feature bags, where RGB captures appearance cues and optical flow captures motion cues. Audio features are not used in the final implementation.

Each video is represented as a fixed-length temporal bag of 32 segments for training and quantitative evaluation. Each segment is represented by a 512-dimensional feature vector. Train-fitted normalisation is applied so that validation and test data do not influence the training statistics.

B. Weakly Supervised MIL

Each video is represented as a bag:

$$B = \{x_1, x_2, \dots, x_T\}, \quad (1)$$

where $T = 32$ and $x_t \in \mathbb{R}^{512}$ is the feature vector for segment t . The video-level label is $y \in \{0, 1\}$, where 0 denotes normal and 1 denotes anomaly. Segment-level labels are unknown.

The model predicts a logit for each segment:

$$s_t = f_{\theta}(x_t). \quad (2)$$

The video-level anomaly score is computed using top- K pooling:

$$S_{\text{video}} = \frac{1}{K} \sum_{t \in \text{TopK}(s; K)} s_t \quad (3)$$

where $K = 4$ in the final configuration. Top- K pooling is used because it is less sensitive to one noisy segment than max pooling.

C. Baseline MIL Model

The baseline model applies a shared segment-level multi-layer perceptron independently to each segment:

$$512 \rightarrow 256 \rightarrow 1. \quad (4)$$

This produces 32 segment logits per video, followed by top- K pooling. The baseline has 263,169 trainable parameters. Its main limitation is that each segment is scored independently, so the model does not explicitly compare a segment with surrounding temporal context.

D. Temporal Attention MIL Model

The Attention MIL model adds temporal self-attention over the segment sequence. The architecture first projects segment features to a hidden representation and then applies Transformer encoder layers with multi-head self-attention. The final scorer maps each contextualised segment representation to an anomaly logit.

The attention model uses two Transformer encoder layers, four attention heads, a feed-forward dimension of 1024 and dropout 0.3. It has 1,743,873 trainable parameters. This is temporal attention over segment features, not spatial attention over raw image frames.

E. Loss Function

Training follows the MIL ranking principle:

$$L_{rank} = \max(0, 1 - S_{video}^a + S_{video}^n), \quad (5)$$

where S_{video}^a is the predicted score for an anomalous bag and S_{video}^n is the predicted score for a normal bag.

A sparsity term discourages assigning high anomaly scores to every segment:

$$L_{sparse} = \sum_{t=1} \sigma(s_t). \quad (6)$$

The final loss is:

$$L = L_{rank} + \lambda_{sp} L_{sparse} \quad (7)$$

where $\lambda_{sp} = 8 \times 10^{-5}$.

F. CLIP Semantic Extension

The extra-credit CLIP semantic extension uses a frozen CLIP ViT-B/32 text encoder to embed 15 anomaly labels: normal, abuse, arrest, arson, assault, burglary, explosion, fighting, road accident, robbery, shooting, shoplifting, stealing, vandalism and violence.

The Attention MIL model is extended with a semantic projection head that maps the pooled video representation into CLIP text space. The total loss is:

$$L_{total} = L_{rank} + \lambda_{sp} L_{sparse} + 0.1 L_{sem}. \quad (8)$$

Only the semantic projection and logit-scale parameter are trained; the CLIP text encoder remains frozen.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

The final configuration uses Adam optimisation with learning rate 1×10^{-5} , batch size 4, early stopping patience 5 and seed 42. Video-level ROC-AUC and Average Precision (AP) are the primary metrics. Segment-expanded AUC is not treated as official frame-level AUC because the model does not use frame-level temporal ground truth.

TABLE III: Final Experimental Configuration

Setting	Value
Feature mode	3D / spatiotemporal
Segments per video	32 for quantitative evaluation
Feature dimension	512
Pooling	Top- K , $K = 4$
Optimiser	Adam
Learning rate	1×10^{-5}
Batch size	4
Max epochs / patience	40 / 5
Ranking margin	1.0
Sparsity weight	8×10^{-5}
CLIP semantic loss weight	0.1
Seed	42

TABLE IV: Main Results Under Top- K Pooling

Model	Val AUC	Val AP	Test AUC	Test AP
Baseline MIL	0.9319	0.9281	0.9312	0.9534
Attention MIL	0.9507	0.9455	0.9498	0.9630
Attention MIL + CLIP	0.9910	0.9878	0.9765	0.9846

B. Experiment 1: Baseline MIL

The first experiment implements the Sultani-style MIL baseline. This establishes whether a simple segment-level MLP with ranking loss and top- K pooling can separate normal and anomalous videos using only video-level labels. The baseline achieves validation AUC/AP of 0.9319/0.9281 and test AUC/AP of 0.9312/0.9534. This provides a strong reference point for later improvements.

C. Experiment 2: Temporal Attention MIL

The second experiment adds temporal self-attention to the MIL model. The purpose is to test whether modelling segment-to-segment relationships improves anomaly ranking compared with independent segment scoring. Attention MIL improves validation AUC/AP to 0.9507/0.9455 and test AUC/AP to 0.9498/0.9630. Table IV and Table V show that attention improves both validation and test results over the baseline.

D. Experiment 3: Pooling and Hyperparameter Ablations

The third experiment studies design choices that affect MIL stability. Top- K pooling is compared with max pooling, and different feature and learning-rate settings are considered. Top- K pooling consistently matches or exceeds max pooling. For the baseline, test AUC is 0.9312 with top- K compared with 0.9274 using max pooling. For Attention MIL, test AUC is 0.9498 with top- K compared with 0.9485 using max pooling.

A higher learning rate of 1×10^{-4} was less stable and dropped attention performance in exploratory runs. The final setting of 1×10^{-5} and $K = 4$ gave the best overall balance between stability and performance. The early 2D feature run achieved a high numerical AUC, but 3D spatiotemporal features were retained because motion modelling is more appropriate for VAD.

TABLE V: Attention MIL Improvement Over Baseline

Split	Base AUC	Att. AUC	Δ AUC	Base AP	Att. AP	Δ AP
Val	0.9319	0.9507	+0.0188	0.9281	0.9455	+0.0174
Test	0.9312	0.9498	+0.0186	0.9534	0.9630	+0.0096

TABLE VI: Hyperparameter Exploration for Attention MIL

Configuration	Test AUC
2D feature run	0.9639*
3D, LR= 1×10^{-4} , $K = 8$	≈ 0.9053
3D, LR= 1×10^{-5} , $K = 8$	≈ 0.9509
3D, LR= 1×10^{-5} , $K = 4$ final	0.9498

*Higher numerical AUC, but less appropriate for VAD because 2D features weakly model motion.

E. Training Behaviour and Validation PR Curves

Fig. 1 shows that all models converge steadily. The baseline decreases gradually, while Attention MIL drops faster and reaches a lower MIL loss. The CLIP semantic model continues training for more epochs because it optimises both MIL ranking and semantic classification objectives.

Fig. 2 compares validation PR curves. Attention MIL achieves higher validation AP than the baseline. These PR curves are validation diagnostics; the official held-out test results are reported in Table IV.

F. Experiment 4: CLIP Semantic Extension

The fourth experiment adds a CLIP-based semantic projection head to the Attention MIL model. The purpose is not only to improve binary anomaly ranking, but also to test whether the model can produce interpretable text-label suggestions. The integrated CLIP semantic model achieves the highest binary video-level metrics, with test AUC/AP of 0.9765/0.9846. However, its fine-grained semantic macro-F1 remains low at 0.2390, meaning it should be treated as a semantic interpretability extension rather than a reliable 15-way anomaly classifier.

Fig. 3 shows the normalised 15-way CLIP semantic confusion matrix. This matrix is a semantic diagnostic over expanded segment-level predictions, while AUC/AP are reported at video level. The model recognises dominant labels such as normal and violence more reliably, but minority anomaly categories remain confused with normal or broad anomaly labels.

G. Qualitative Evaluation

Fig. 4 compares raw video-level mean features with model-guided top- K segment features. The mean-feature UMAP shows strong overlap between normal and anomaly points. This is expected because anomalous videos often contain many normal segments, so averaging the full video dilutes anomaly information. The top- K UMAP is more structured because it visualises the segments most responsible for the MIL decision.

Fig. 5 shows segment-level anomaly score curves for a correct normal example, a correct anomaly example and a false-positive failure case. The quantitative evaluation uses fixed

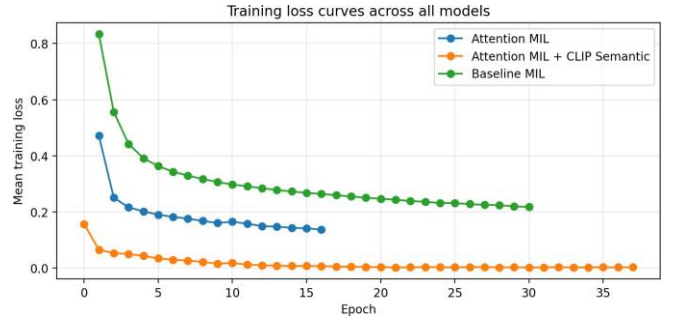


Fig. 1: Training loss curves across Baseline MIL, Attention MIL and Attention MIL + CLIP semantic models. Baseline MIL decreases gradually, Attention MIL converges faster, and the CLIP semantic model trains longer because it optimises both MIL ranking and semantic objectives.

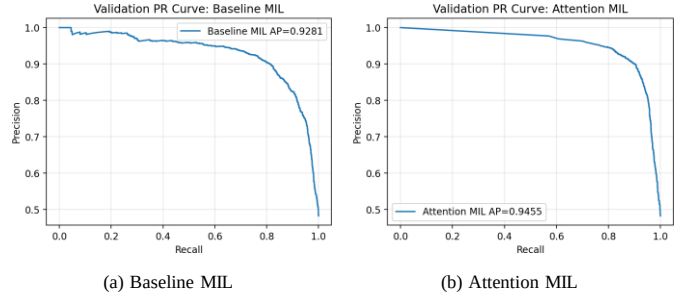


Fig. 2: Validation precision-recall curves for Baseline MIL and Attention MIL. Attention MIL achieves higher validation AP, supporting the quantitative improvement over the baseline.

32-segment bags; some qualitative diagnostic plots may show the available segment-score resolution for selected clips to make temporal behaviour easier to inspect. The false-positive example supports the limitation that weakly supervised MIL can learn context shortcuts from unusual motion or scene changes.

H. Efficiency Analysis

The post-feature MIL heads are lightweight. The baseline model requires approximately 0.080 ms per bag, while Attention MIL requires approximately 0.359 ms per bag. Attention is slower because of the Transformer encoder, but both models are comfortably real-time after feature extraction. The main deployment bottleneck would be $R(2+1)D/I3D$ feature extraction rather than the MIL head.

V. DISCUSSION

A. What Worked

The Sultani-style MIL formulation works well with modern spatiotemporal features. The baseline already achieves strong test performance, and the attention model improves both validation and test AUC/AP. This suggests that comparing temporal segments using self-attention helps the model distinguish abnormal snippets from normal context.

Top- K pooling is also effective. It is more stable than pure max pooling because it reduces the influence of a single noisy high-scoring segment. The UMAP results support the need

Fig. 3: Normalised 15-way confusion matrix for the integrated CLIP semantic head. The matrix is used as a semantic diagnostic and shows that dominant labels are recognised more reliably than minority anomaly categories.

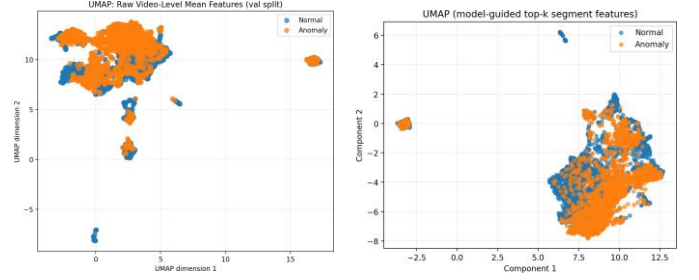
TABLE VII: Integrated CLIP Semantic Head Metrics

Metric	Validation	Test
Binary video AUC	0.9910	0.9765
Binary video AP	0.9878	0.9846
Semantic accuracy	0.9496	0.8872
Semantic macro-F1	0.3631	0.2390
Semantic weighted-F1	0.9495	0.8859

TABLE VIII: Post-Feature Efficiency Comparison

Model	Parameters	Inference (ms/bag)
Baseline MIL	263,169	0.080
Attention MIL	1,743,873	0.359

for MIL: raw mean-pooled video features are not cleanly separable, but model-selected top- K features show more useful structure.



(a) Raw video-level mean features (b) Model-guided top- K features

Fig. 4: UMAP visualisation of feature space. Raw video-level mean features remain mixed because weak video-level anomaly labels contain many normal temporal segments. Model-guided top- K segment features are more structured because they focus on the segments used by the MIL decision.

B. What Did Not Work

The integrated CLIP semantic head improves binary anomaly metrics but does not solve fine-grained anomaly classification. The low macro-F1 shows that many minority classes are not recognised reliably. This is expected because the model is trained mainly from weak video-level labels and because the class distribution is highly imbalanced.

The attention model also produces false positives on some

Segment-Level Anomaly Score Curves - Validation Set (Attention_MIL)

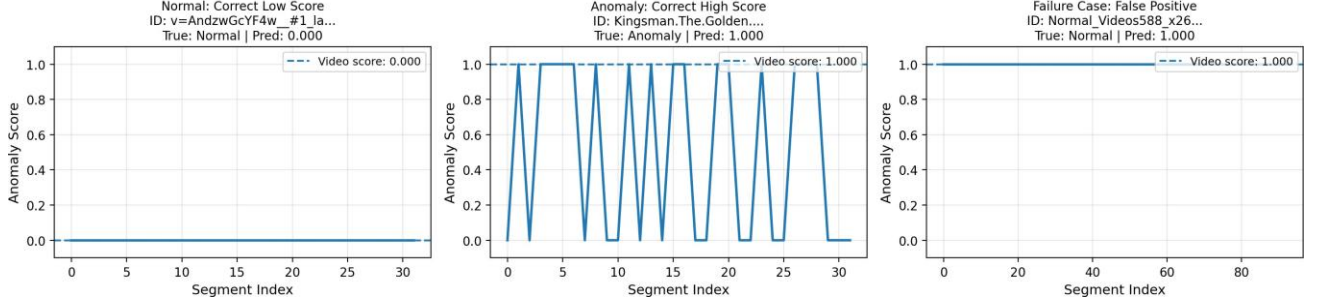


Fig. 5: Segment-level anomaly score curves on the validation set using Attention MIL. The figure shows a correct normal example with low scores, a correct anomalous example with high segment scores, and a false-positive failure case where a normal clip receives a high anomaly score.

normal videos. These cases likely occur when normal videos contain unusual movement, scene changes or visual context that resembles anomaly patterns. This matches the context-bias limitation described in weakly supervised MIL literature.

C. Limitations

The main limitations are:

- Frame-level temporal ground truth is not used, so official frame-level localisation AUC is not reported.
- XD-Violence audio is not used, even though audio can help detect explosions, fighting and other violent events.
- UCF-Crime and XD-Violence use different feature backbones, so direct dataset-to-dataset comparison is limited.
- Fine-grained semantic classification remains weak for minority classes.
- Some normal videos with unusual motion are false positives.

VI. CONCLUSION AND FUTURE WORK

This project demonstrates that weakly supervised MIL is effective for video anomaly detection when combined with strong spatiotemporal feature bags. The baseline MIL model achieves test AUC/AP of 0.9312/0.9534, while temporal attention improves the result to 0.9498/0.9630. The CLIP semantic extension further improves binary ranking to 0.9765/0.9846, but low semantic macro-F1 shows that fine-grained anomaly labelling remains difficult.

The best practical model is therefore Attention MIL for binary anomaly detection. The CLIP semantic head is useful as an interpretability and semantic-labelling extension, but it should not be presented as a fully reliable anomaly-category classifier.

Future work should include full frame-level evaluation using temporal ground truth, unified feature extraction across UCF-Crime and XD-Violence, audio integration for XD-Violence,

class-balanced semantic losses, focal loss, prompt engineering for CLIP labels and UMIL-style unbiased training to reduce context-shortcut bias.

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APPENDIX A ADDITIONAL DIAGNOSTIC FIGURES

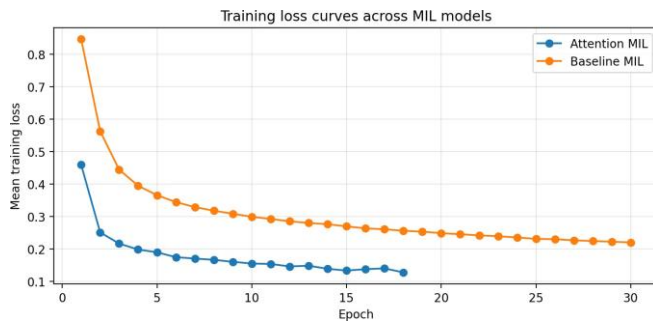


Fig. 6: Training loss curves for the two MIL models only, excluding the CLIP semantic model.

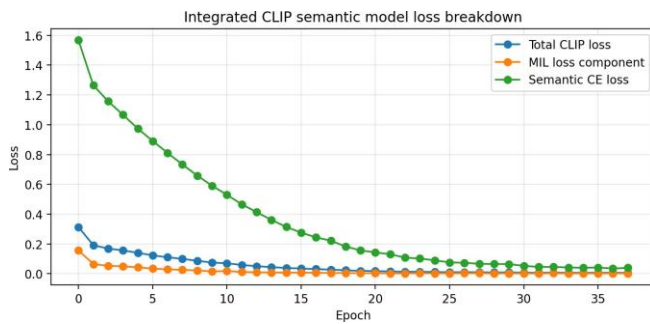


Fig. 7: Integrated CLIP semantic model loss breakdown. The semantic cross-entropy component decreases more slowly than the MIL ranking component.

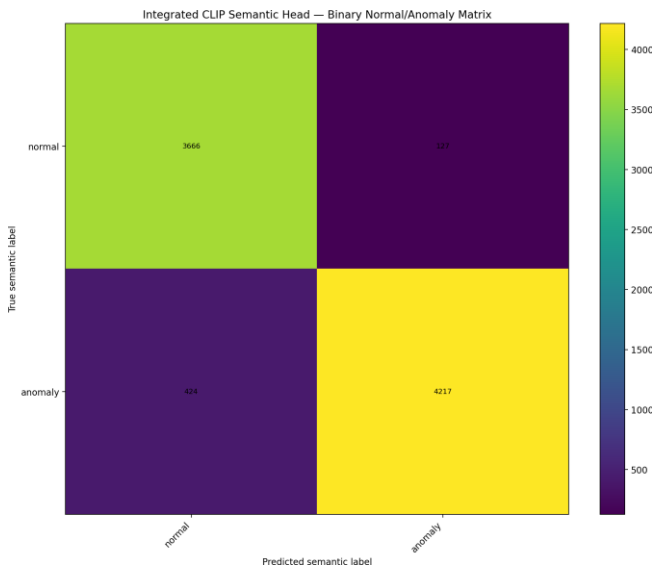


Fig. 8: Binary normal/anomaly confusion matrix for the integrated CLIP semantic head.

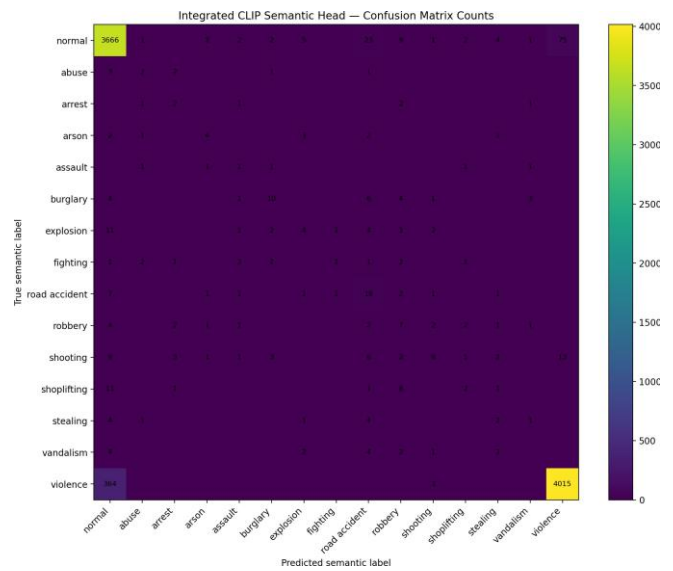


Fig. 9: Raw-count 15-way confusion matrix for the integrated CLIP semantic head. The large normal and violence counts explain why weighted metrics are stronger than macro-F1.

CLIP Text Output — Binary Normal/Anomaly Confusion Matrix

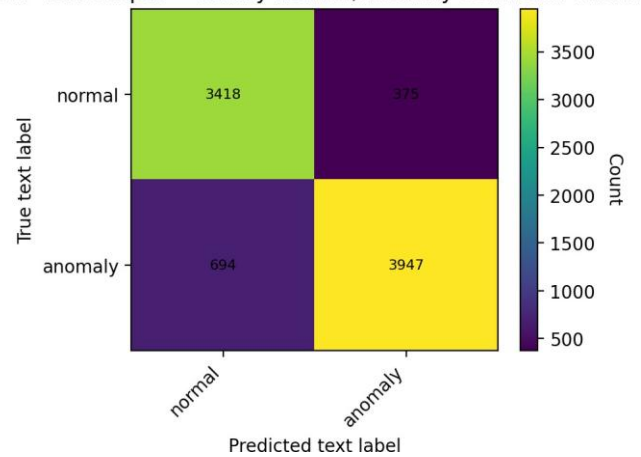


Fig. 10: Binary normal/anomaly confusion matrix for the post-hoc CLIP text-output baseline.

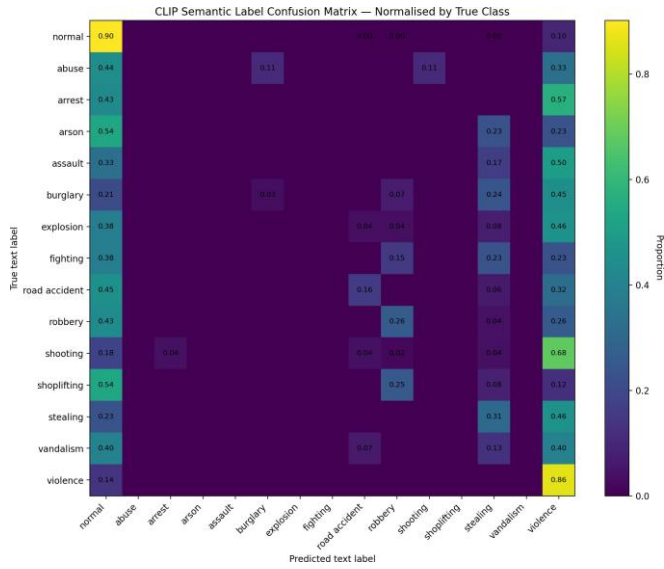


Fig. 11: Normalised confusion matrix for the post-hoc CLIP text-output baseline.

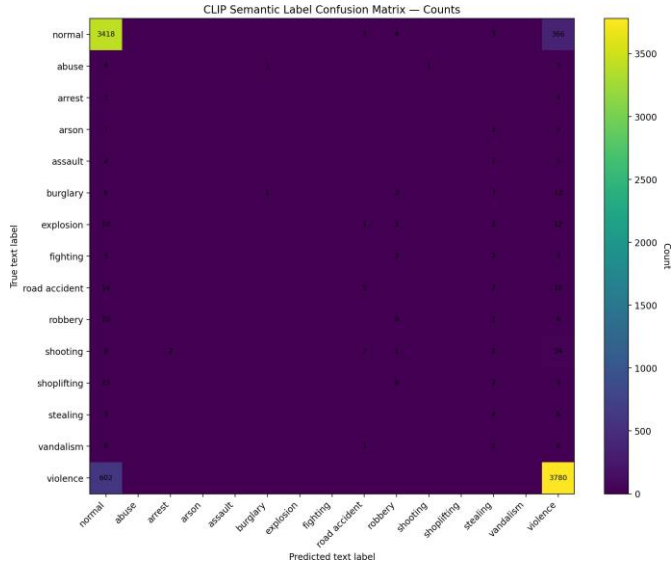


Fig. 12: Raw-count confusion matrix for the post-hoc CLIP text-output baseline.

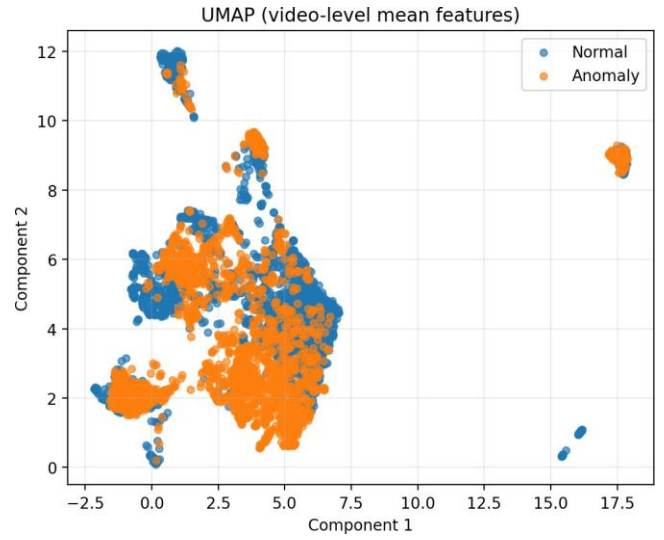


Fig. 13: Additional UMAP visualisation of video-level mean features. Normal and anomalous samples remain mixed, which is expected because video-level labels do not identify which temporal segments are anomalous.

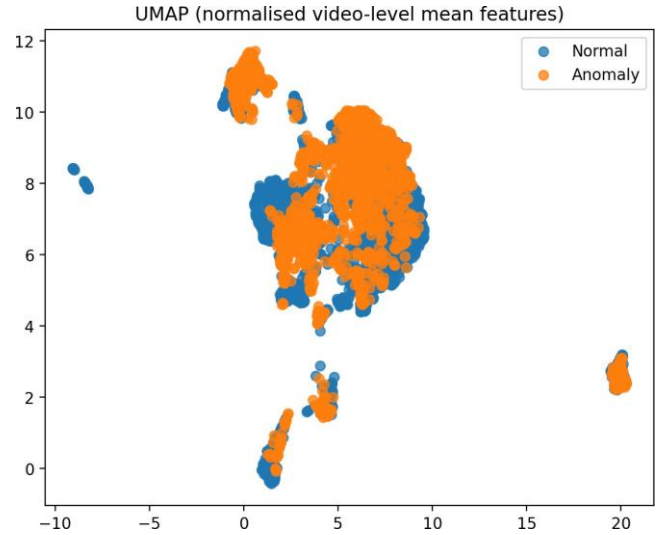


Fig. 14: UMAP of normalised video-level mean features. Normalisation changes the feature geometry, but normal/anomaly overlap remains because the task is weakly supervised.

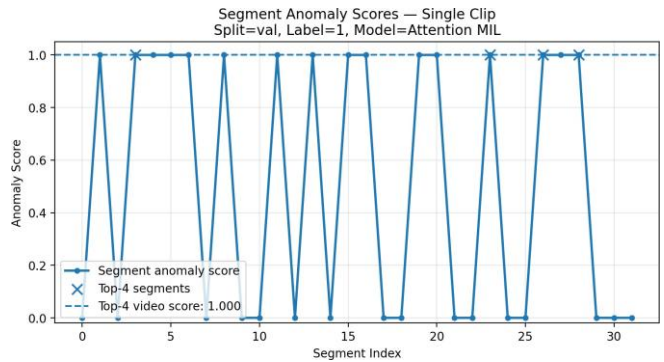


Fig. 15: Single-clip segment anomaly score curve showing the top-4 selected segments used for top- K video-level scoring.

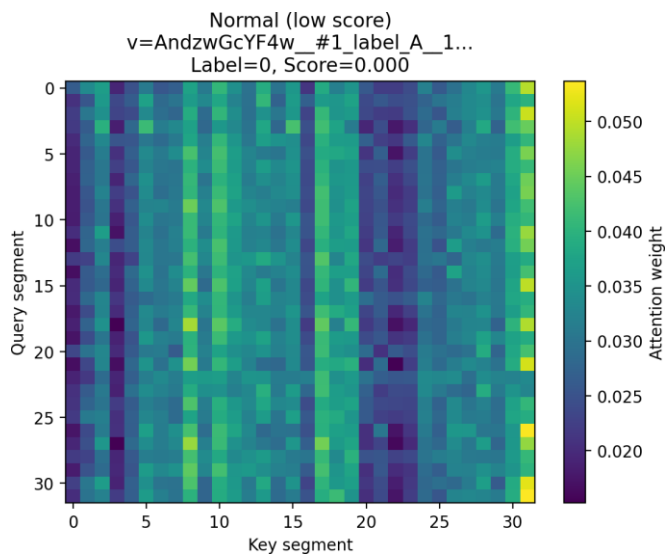


Fig. 16: Attention heatmap for a correctly classified normal low-score clip.

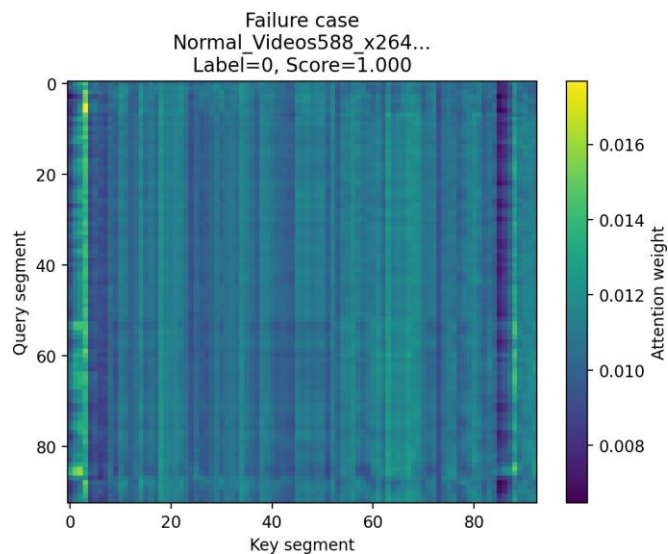


Fig. 18: Attention heatmap for a false-positive failure case. The broad attention pattern supports the interpretation that the model can overreact to unusual but non-anomalous temporal context.

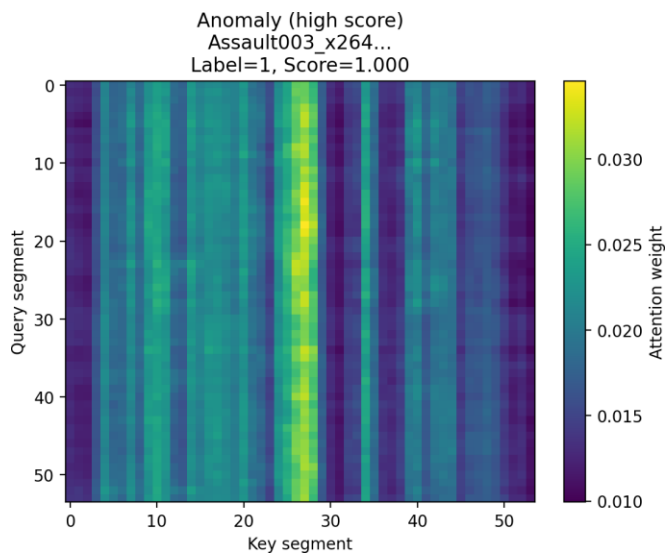


Fig. 17: Attention heatmap for a correctly classified anomalous high-score clip.